Design Rules Summary

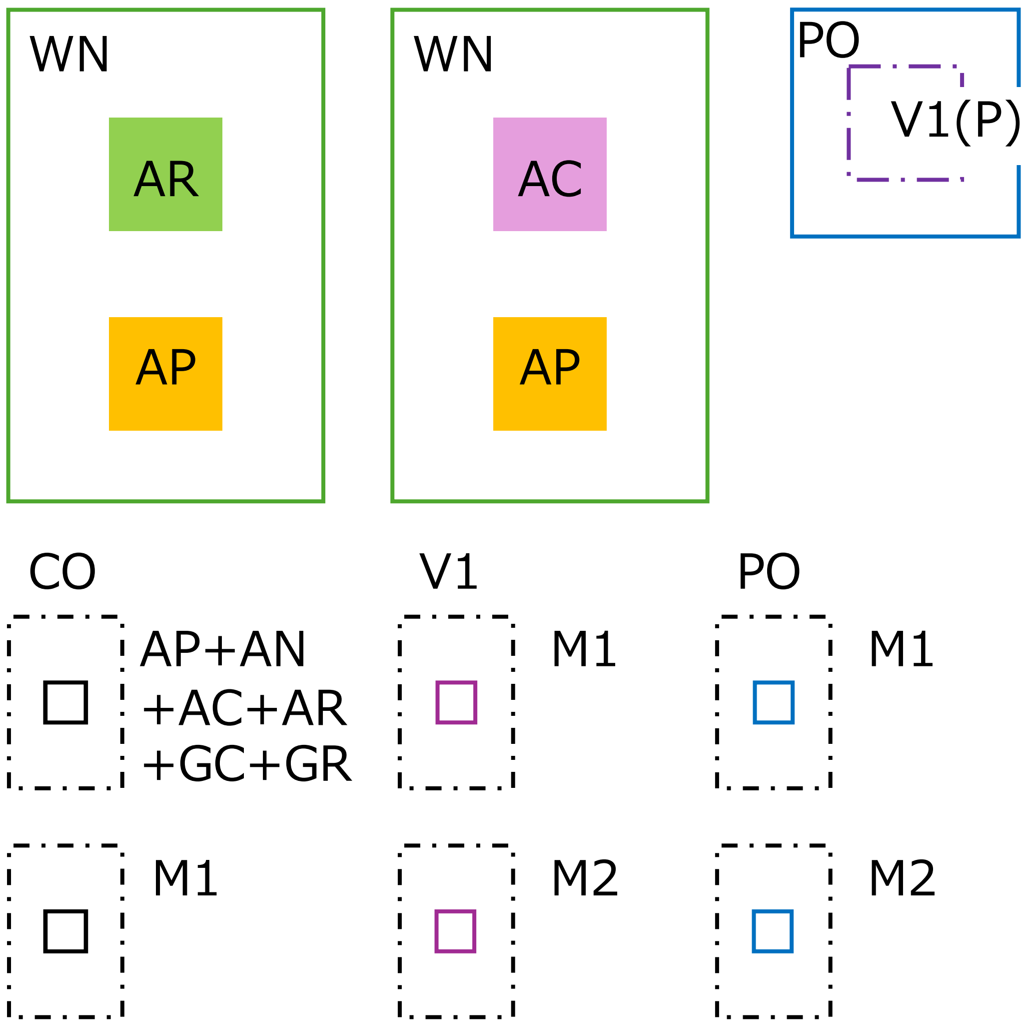
for Drawing Layers

2026/01/14

Drawing Layer Table

	Original MASK layers				Proposed Drawing layers			
	Name	NUM	TYPE		Name	NUM	TYPE	
1	PSUB	140	0		WN	140	0	
2	NW	36	0					MDP generatioed
3	HVNW	141	0					MDP generatioed
4	L	3	0					MDP generatioed
					AP	3	1	Active for P+
					AN	3	2	Active for N+
					AR	3	3	Active for resistor
					AC	3	4	Active for capacitance
5	NF	25	0					MDP generatioed
6	PF	26	0					MDP generatioed
7	CL	143	0					MDP generatioed
8	HPBE	144	0					No USE
9	HNBE	145	0					No USE
10	PBE	146	0					MDP generatioed
11	NBE	147	0					MDP generatioed
12	HPM	33	0					MDP generatioed
13	RHP	148	0					No USE
14	SG	8	0					
					GC	8	1	Gate Poly transistor
					GR	8	2	Gate POLY resistor
15	PM	35	0					MDP generatioed
16	NM	7	0					MDP generatioed
17	R	12	0					MDP generatioed
18	PSD	9	0					MDP generatioed
19	NSD	28	0					MDP generatioed
20	CONT	11	0		CO	11	0	
21	M1	13	0		M1	13	0	
22	TC	19	0		V1	19	0	
23	M2	20	0		M2	20	0	
24	PRO	14	0		PO	14	0	
	TXM1	48	0		TXM1	48	0	M1 Label
	PIN	48	1		PIN	48	1	M1 Pin
	TXM2	49	0		TXM2	49	0	M2 Label
	MASK	63	0		MASK	63	0	Waiver region
	MEAS	63	1		MEAS	63	1	Measurement region
	SCRIBE	80	0		SCRIBE	80	0	Waiver region (EDGE SEAL)
	prBoundary	235	0		prBoundary	235	0	Cell boundary

Prohibited



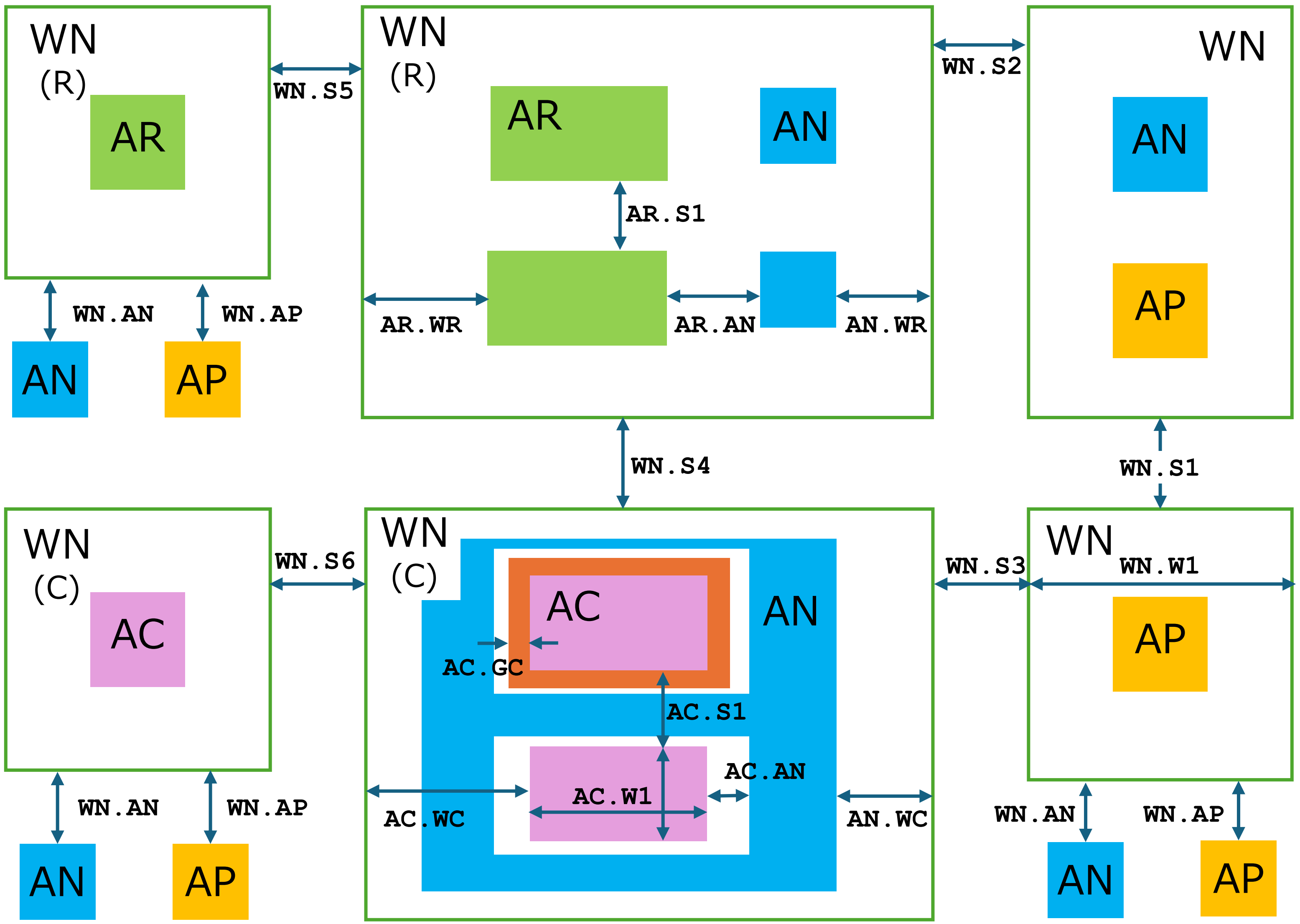
Rule	L1	L2	Func	MIN	MAX
WN.Z1	WN(R)	AP+AC	None	0.0	
WN.Z2	WN(C)	AP+AR	None	0.0	
CO.Z1	CO	AA+GC+GR	None	0.0	
CO.Z3	CO	M1	None	0.0	
V1.Z1	V1	M1	None	0.0	
V1.Z2	V1	M2	None	0.0	
PO.Z1	PO	M1	None	0.0	
PO.Z2	PO	M2	None	0.0	
PO.Z3	PO	V1(P)	Exist	0.0	

Function:

Wmin/max: L1 Width min/max
Smin/max: L1 Space min/max (L1=L2)
Smin/max: L1 Separate L2 min/max
Emin/max: L1 Enclosed L2 min/max
Fmin/max: L1 Enclosing L2 min/max
ECmin: MOS Endcap min
Wfix: L1 Windth equal
Sfix: Space equal
Donut: L1 Souuounded by L2
None: Not Exist

V1(P):V1 on PE

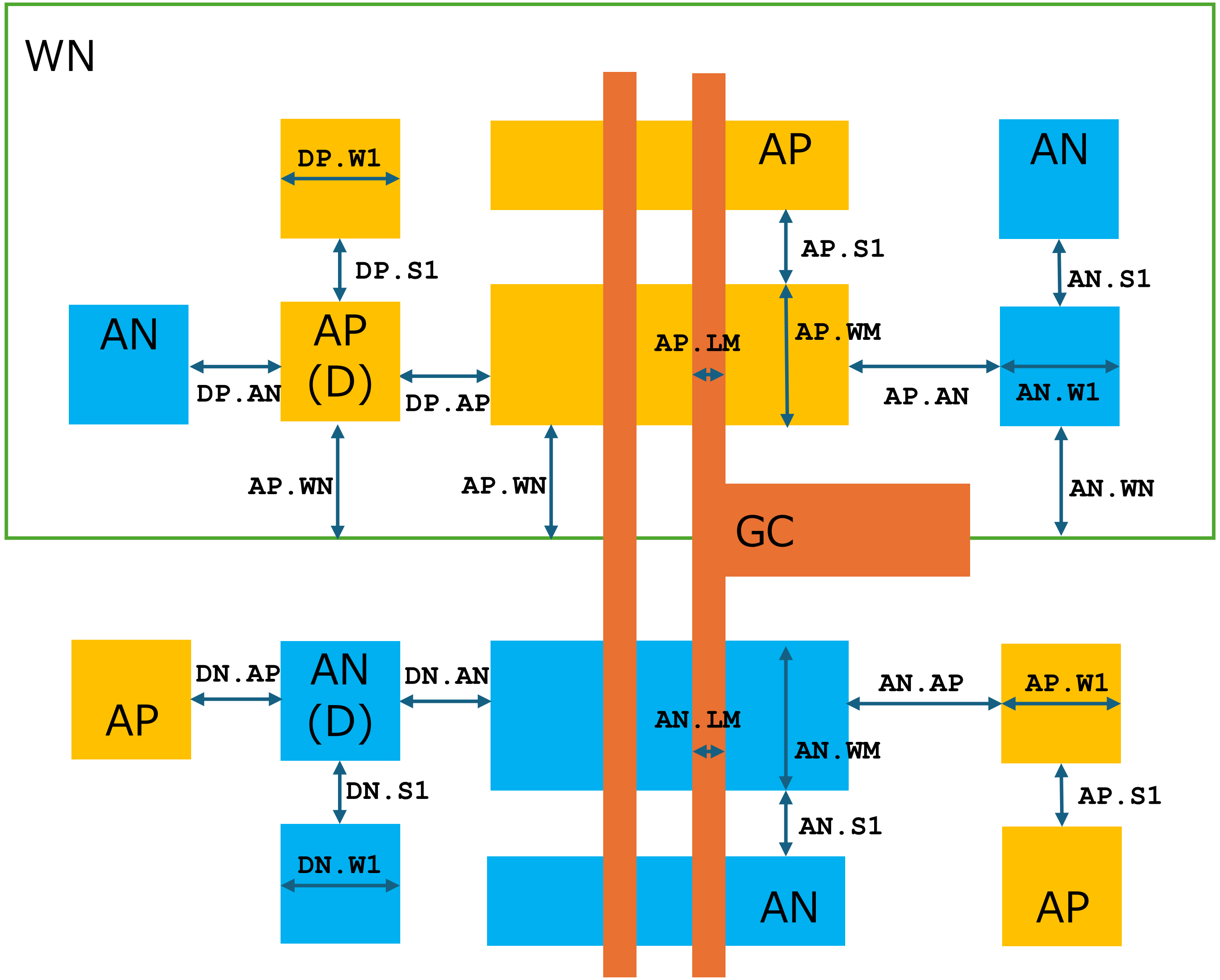
WN to AP/AN/AR/AC related



Rule	L1	L2	Func	MIN	MAX
WN.W1	WN		Wmin	8.0	
WN.S1	WN	WN	Smin	12.0	
WN.S2	WN	WN(R)	Smin	9.5	
WN.S3	WN	WN(C)	Smin	12.0	
WN.S4	WN(R)	WN(C)	Smin	9.5	
WN.S5	WN(R)	WN(R)	Smin	8.0	
WN.S6	WN(C)	WN(C)	Smin	12.0	
WN.AP	WN	AP	Smin	5.0	
WN.AN	WN	AN	Smin	10.0	
AR.S1	AR	AR	Smin	4.0	
AR.WR	AR	WN(R)	Emin	10.0	
AN.WR	AN	WN(R)	Emin	5.0	
AR.AN	AR	AN	Smin	4.0	
AC.W1	AC		Wmin/max	28.5	120.0
AC.S1	AC	AC	Smin	???	???
AC.WC	AC	WN(C)	Emin	???	???
AN.WC	AN	WN(C)	Emin	3.0	
AC.AN	AC	AN	Smin	2.8	
AC.GC	AC	GC	Emin	1.4	
AC.R1	AC	AN(C)	Donut	0.0	

WN(R) : WN overlapped AR
WN(C) : WN overlapped AC
AN(C) : AN in WN(C)

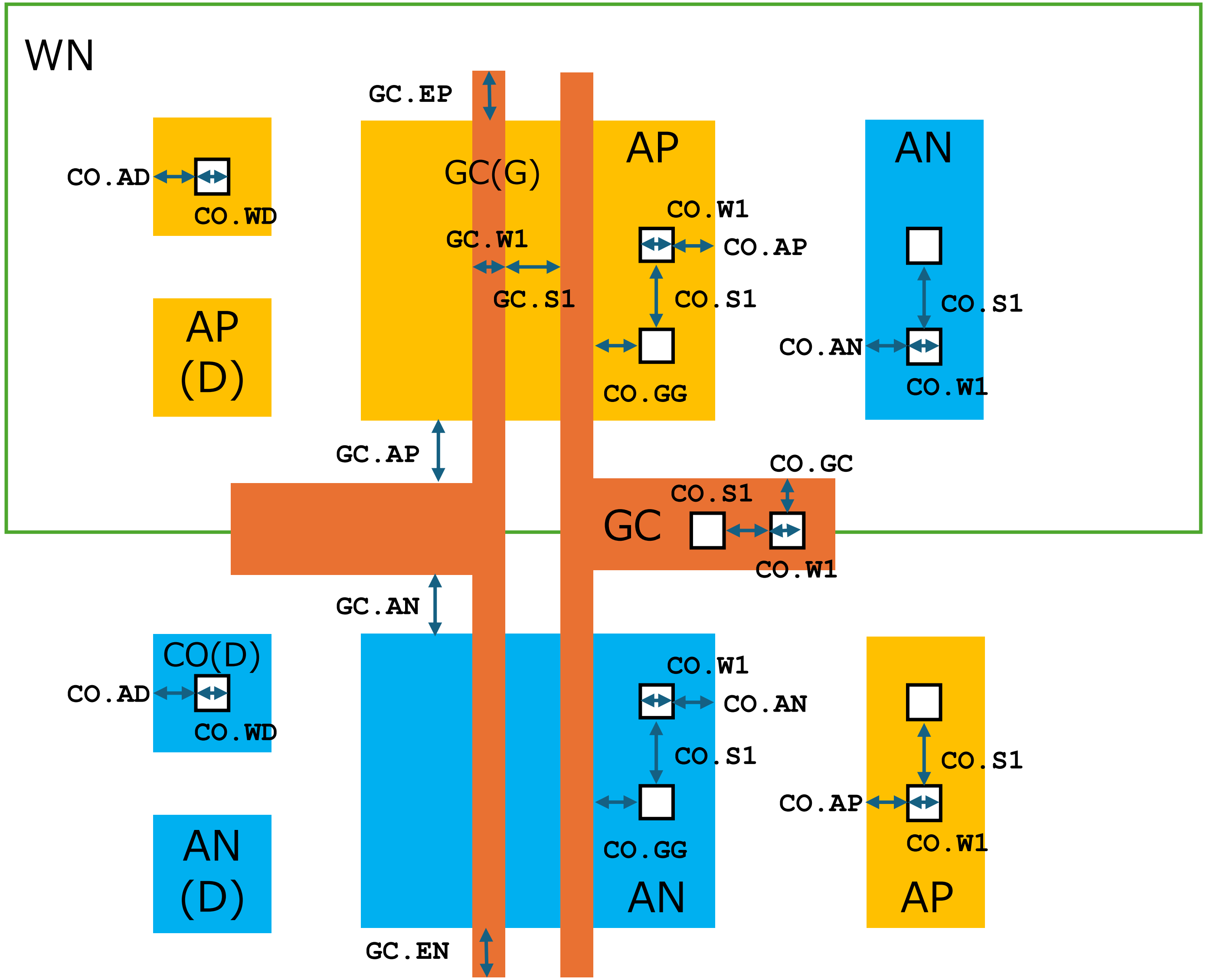
MOS transistor related



Rule	L1	L2	Func	MIN	MAX
AP.W1	AP		Wmin	1.4	
AP.S1	AP	AP	Smin	1.4	
AP.AN	AP	AN	Smin	2.8	
AP.WM	PMOS	AP-GC	Wmin/max	3.4	60.0
AP.LM	PMOS	GC-AN	Lmin/max	1.0	30.0
AP.WN	AP	WN	Emin	7.0	
DP.W1	DP		Wmin	2.8	
DP.S1	DP	DP	Smin	???	???
DP.AP	DP	AP	Smin	2.8	
DP.AN	DP	AN	Smin	2.8	
AN.W1	AN		Wmin	1.4	
AN.S1	AN	AN	Smin	1.4	
AN.AP	AN	AP	Smin	2.8	
AN.WM	NMOS	AN-GC	Wmin/max	3.4	60.0
AN.LM	NMOS	GC-AN	Lmin/max	1.0	30.0
AN.WN	AN	WN(M)	Emin	5.0	
DN.W1	DN		Wmin	2.8	
DN.S1	DN	DN	Smin	???	???
DN.AP	DN	AP	Smin	2.8	
DN.AN	DN	AN	Smin	2.8	

WN(M)：WN overlapped AN/AP

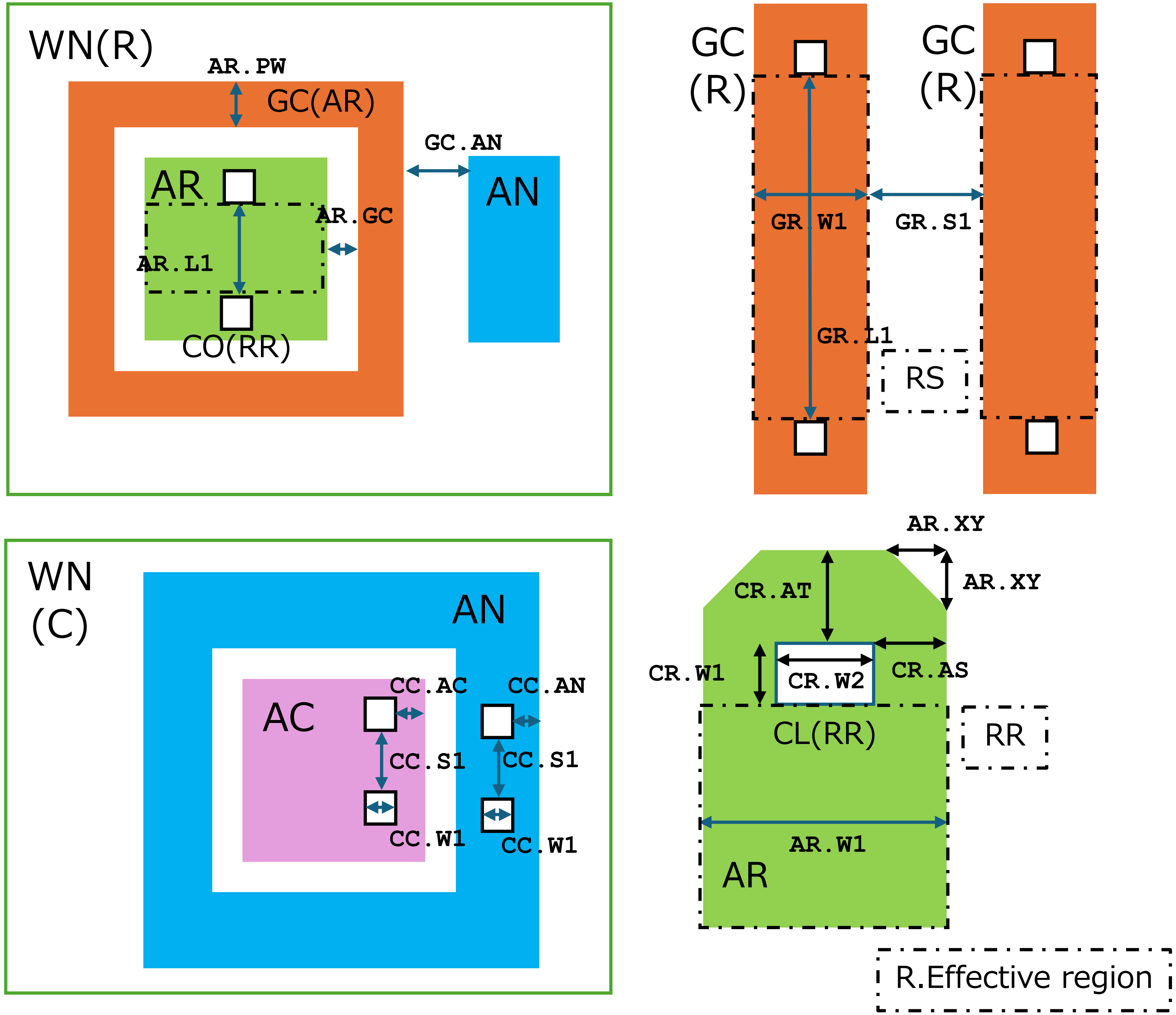
Contact and Poly related



Rule	L1	L2	Func	MIN	MAX
GC.W1	GC		Wmin	1.0	30.0
GC.S1	GC	GC	Smin	1.2	
GC.AP	GC	AP	Smin	0.4	
GC.AN	GC	AN	Smin	0.4	
GC.EP	PMOS	GC	ECmin	1.2	
GC.EN	NMOS	GC	ECmin	1.2	
CO.W1	CO		Wmin	1.0	
CO.S1	CO	CO	Smin	1.0	
CO.WD	CO(D)		Wfix	1.2	
CO.AD	CO(D)	AP+AN	Emin	1.2	
CO.AP	CO	AP	Emin	0.8	
CO.AN	CO	AN	Emin	0.8	
CO.GG	CO	GC(G)	Smin	1.0	
CO.GC	CO	GC+GR	Emin	0.8	

AP(D)：AP not interacted GC
AN(D)：AN not interacted GC
CO(D)：CO on AP(D)/AN(D)
GC(G)：GC for PMOS/NMOS

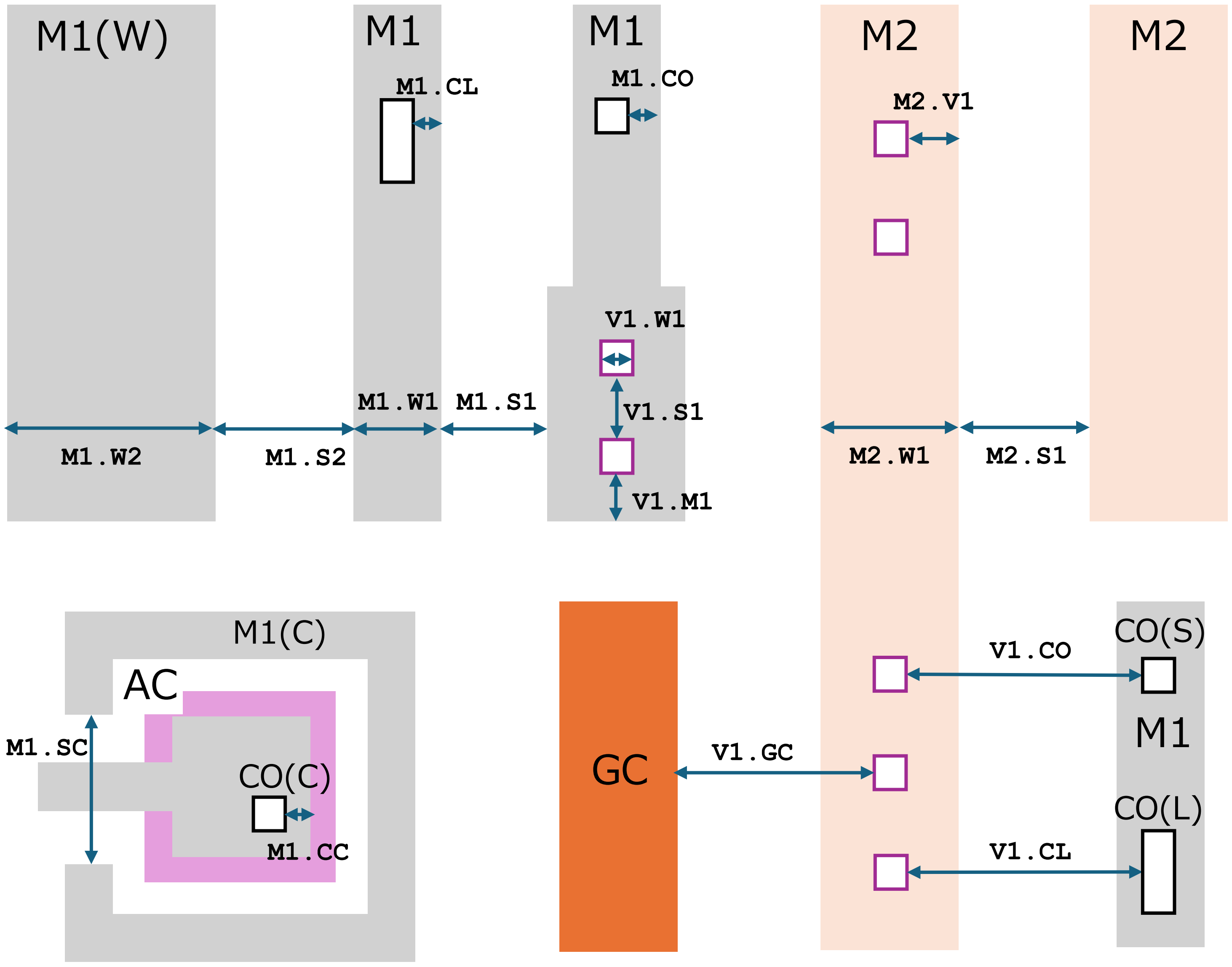
Resistor(AR/RS) and Capacitor(AC) related



Rule	L1	L2	Func	MIN	MAX
AR.W1	RR	RR(L)	Wmin/max	2.8	20.0
AR.L1	RR	RR(W)	Lmin/max	13.0	100.0
AR.GC	GC	AR	Smin	1.0	
AR.PW	GC(AR)		Wmin	2.0	
AR.XY	AR	Bevel	XYmin	1.0	
GC.AN	GC(AR)	AN	Smin	???	???
GC.R1	AR	GC(AR)	Donut	0.0	
GC.R2	GC	TieDown		0.0	
GR.W1	RS	RS(L)	Wmin/max	4.0	20.0
GR.L1	RS	RS(W)	Lmin/max	20.0	100.0
GR.S1	RS	RS	Smin	2.0	
CR.W1	CO(RR)		Wmin	1.0	
CR.W2	CL(RR)		Wmin/max	1.0	97.5
CR.AT	CO(RR)	AR(T)	Emin	1.5	
CR.AS	CO(RR)	AR(S)	Emin	1.0	
CC.W1	CO(C)		Wmin	1.2	
CC.S1	CO(C)	CO(C)	Smin	1.2	
CC.AC	CO(C)	AC	Emin	2.9	
CC.AN	CO(C)	AN	Emin	1.2	

GC(AR) : GC in WN(R)
RS(L) : Length of AR effective
RS(W) : Width of AR effective
CO(RR) : CO short side on AR
CL(RR) : CO large side on AR
AR(T) : AR top fringe
AR(S) : AR side fringe
CO(C) : CO on AC

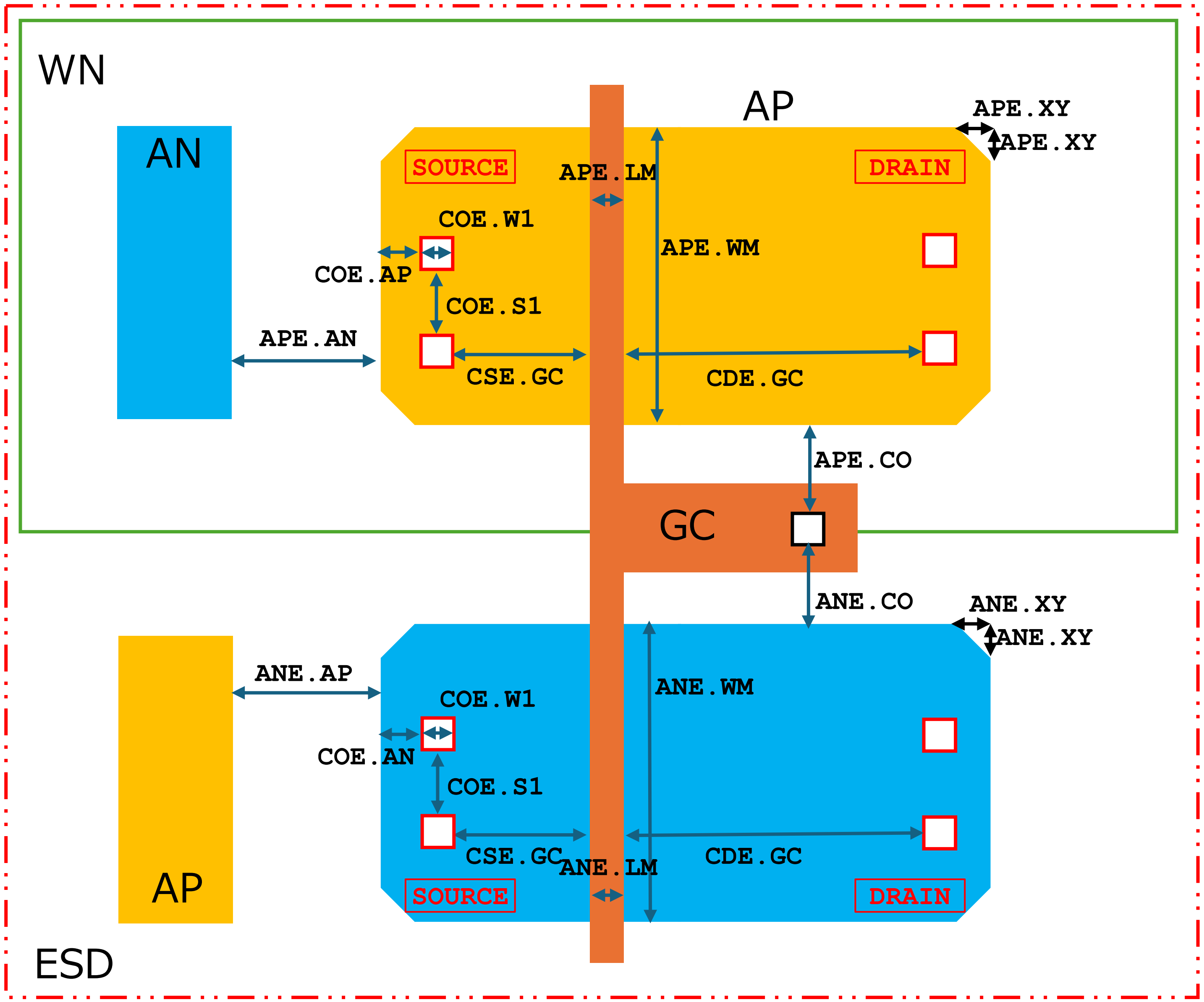
M1/V1/M2 related



Rule	L1	L2	Func	MIN	MAX
M1.W1	M1		Wmin	1.8	
M1.S1	M1	M1	Smin	1.4	
M1.SC	M1(C)	M1(C)	Smin	10.0	
M1.CO	M1	CO(S)	Fmin	0.8	
M1.CC	M1	CO(C)	Fmin	1.2	
M1.CL	M1	CO(L)	Fmin	1.2	
M1.SW	M1(W)	M1	Smin	2.0	
V1.W1	V1		Wfix	1.4	
V1.S1	V1	V1	Smin	1.5	
V1.M1	V1	M1	Emin	1.0	
V1.GC	V1	GC	Smin	1.2	
V1.CO	V1	CO	Smin	1.0	
V1.CL	V1	CO(L)	Smin	1.4	
M2.W1	M2		Wmin	3.0	
M2.S1	M2	M2	Smin	2.0	
M2.V1	M2	V1	Fmin	1.0	

M1(C)：M1 in WN(C)
CO(C)：CO on AC
CO(S)：CO small contact
CO(L)：CO large contact (only on AR)
M1(W)：Wide M1

ESD MOS transistor related

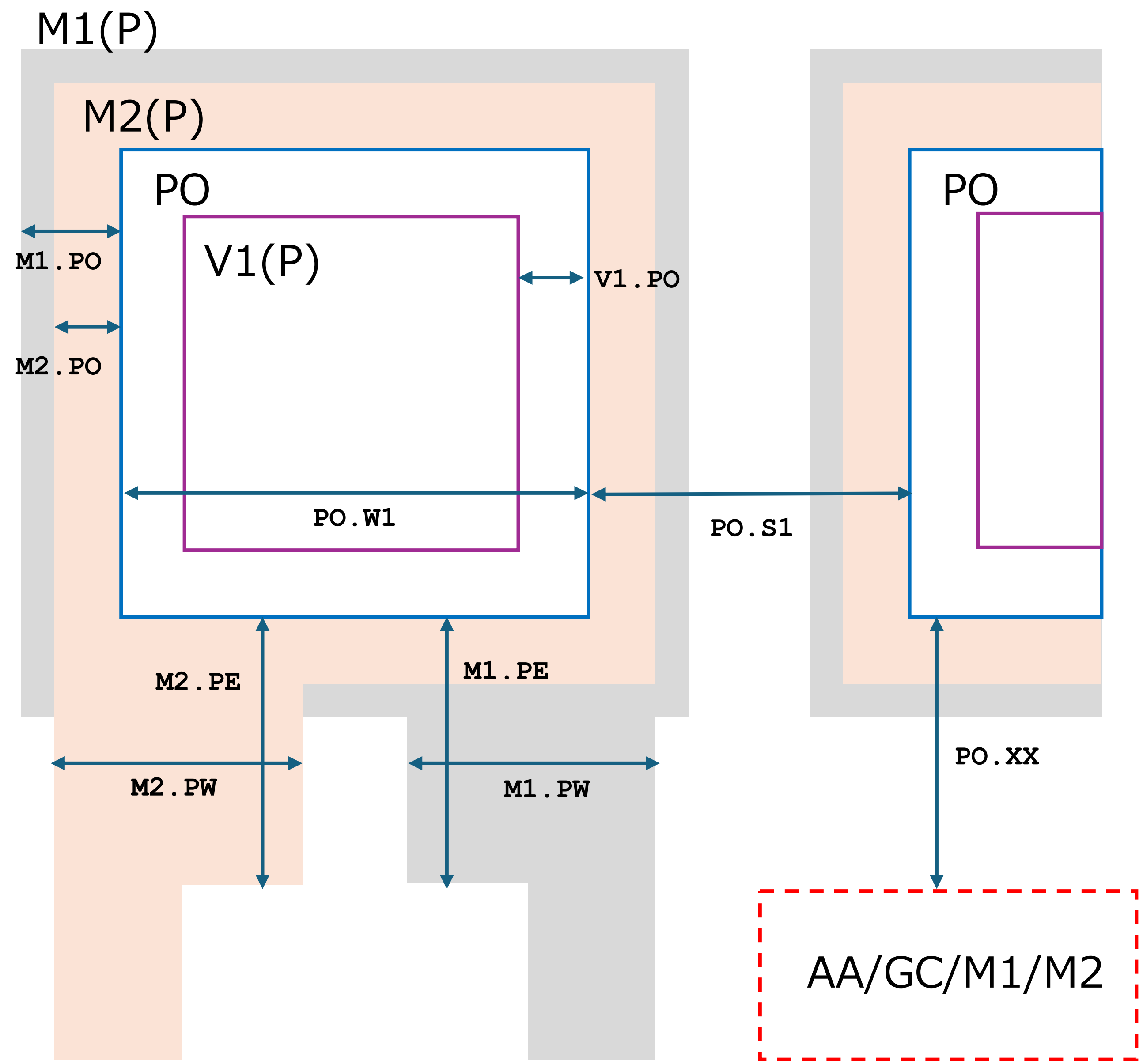


Rule	L1	L2	Func	MIN	MAX
APE.WM	PMOS	AP-GC	Wmin/max	11.0	60.0
APE.LM	PMOS	GC-AN	Lmin/max	2.0	30.0
APE.AN	APE	AN	Smin	10.0	
APE.XY	APE	Bevel	XYmin	1.4	
APE.CO	APE	CO	Smin	2.0	
ANE.WM	NMOS	AN-GC	Wmin/max	11.0	60.0
ANE.LM	NMOS	GC-AN	Lmin/max	2.0	30.0
ANE.AP	ANE	AP	Smin	10.0	
ANE.XY	ANE	Bevel	XYmin	1.4	
ANE.CO	ANE	CO	Smin	2.0	
COE.W1	CO(E)		Wfix	3.0	
COE.S1	CO(E)		Smin	1.6	
CDE.GC	CD(E)	GC	Smin	7.0	
CSE.GC	CS(E)	GC	Smin	3.0	
COE.AP	CO(E)	AP	Emin	4.0	
COE.AN	CO(E)	AN	Emin	4.0	

APE/ANE) : AP/AN on ESD)

CO(E) : CO on ESD

PAD related



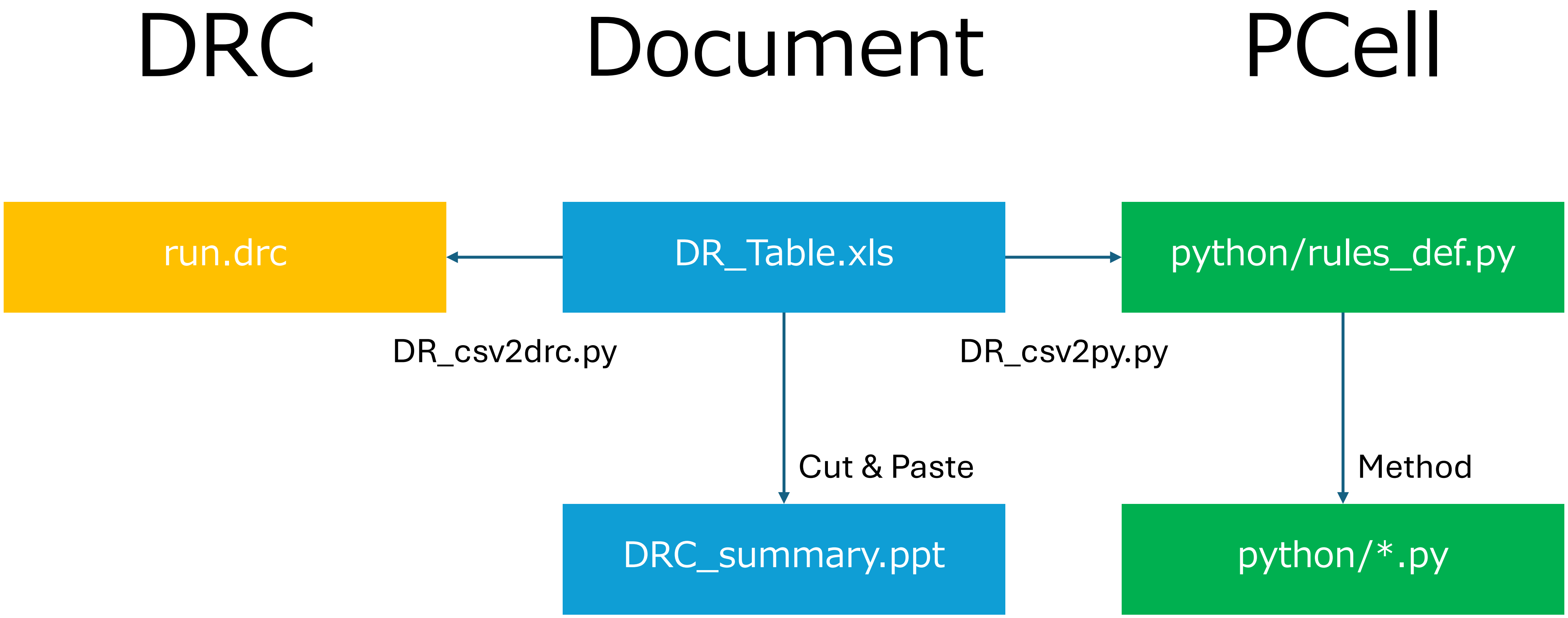
Rule	L1	L2	Func	MIN	MAX
PO.W1	PO		Wmin	70.0	
PO.S1	PO		Smin	64.0	
M2.PO	M2(P)	PO	Fmin	5.0	
M1.PO	M1(P)	PO	Fmin	5.0	
PO.V1	PO	V1(P)	Emin	10.0	
PO.AP	PO	AP	Smin	14.0	
PO.AN	PO	AN	Smin	14.0	
PO.GC	PO	GC	Smin	14.0	
PO.M1	PO	M1	Smin	14.0	
PO.M2	PO	M2	Smin	14.0	
M2.PE	M2	Extention		19.0	
M2.PW	M2	Extention	Wmin	40.0	
M1.PE	M2	Extention		19.0	
M1.PW	M2	Extention	Wmin	40.0	

** Not yet implemented

M1(P) : M1 on PO
M2(P) : M2 on PO
V1(P) : V1 on PO

Design Rule cotrol

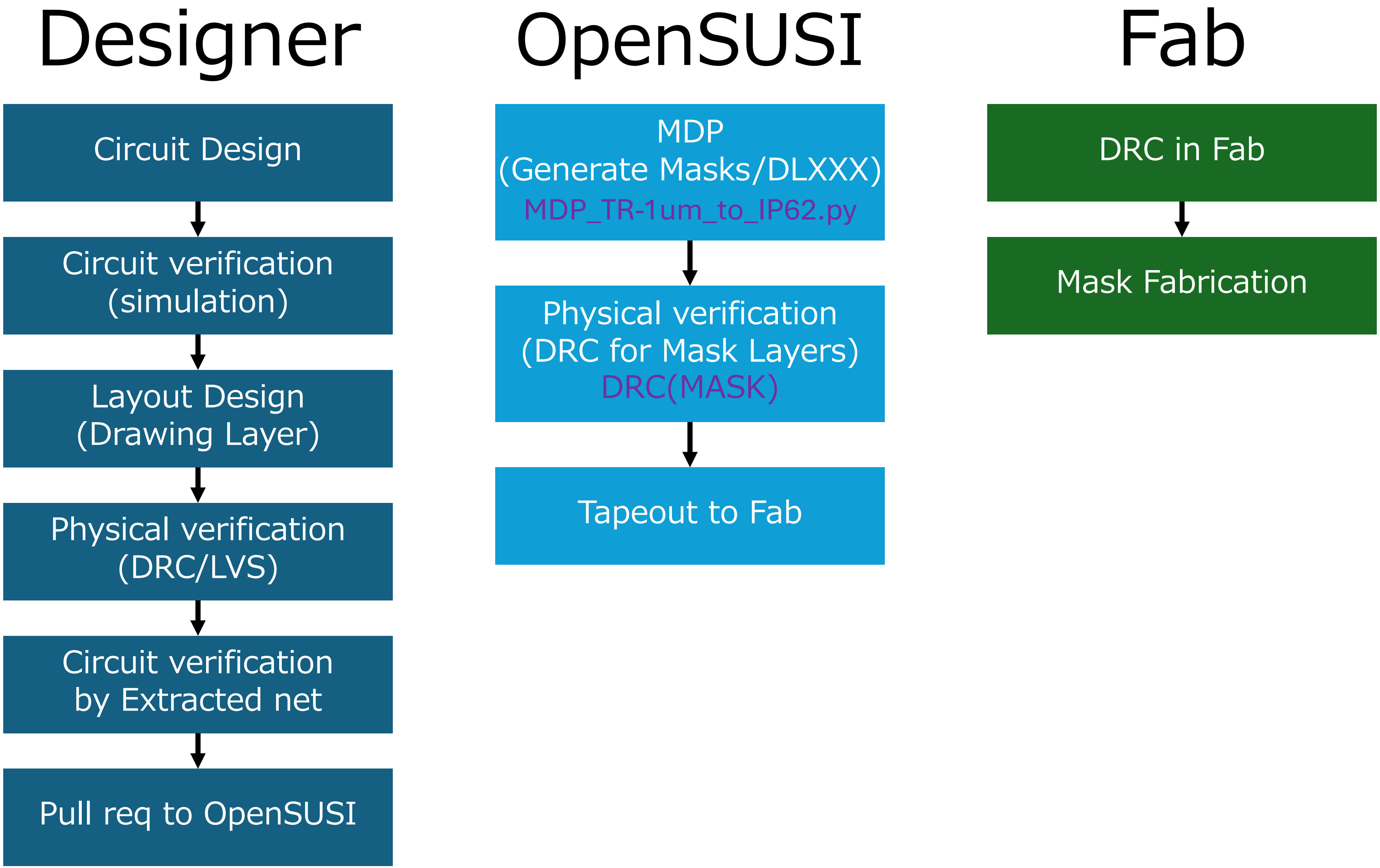
DRC integrity



ALL Critical Dimension parameters are referenced by ‘XX.YY’ rule’s name.

Design to Tapeout Process

Mask Data Preparation


Designer
Circuit Design
Circuit verification (simulation)
Layout Design (Drawing Layer)
Physical verification (DRC/LVS)
Circuit verification by Extracted net
Pull req to OpenSUSI
OpenSUSI
MDP (Generate Masks/DLXXX)
MDP_TR-1um_to_IP62.py
Physical verification (DRC for Mask Layers)
DRC(MASK)
Tapeout to Fab
Fab
DRC in Fab
Mask Fabrication